CHEN Han

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CONTACT INFORMATION

Tel: +86-13305014345
Tel: +65-89420214
Wechat: concyclics
E-mail: chenhan@u.nus.edu
Github: www.github.com/Concyclics
Address: NUS School of Computing, 13 Computing Drive, Singapore 117417

EDUCATION

National University of Singapore, Singapore. 2025–now
- Doctor of Philosophy, Computer Science. Supervisor: Prof.HE Bingsheng

National University of Singapore, Singapore. 2023–2025
- Master of Computing, Computer Science Specialization. GPA: 4.4/5.0

South China University of Technology (SCUT), Guangzhou, China. 2019–2023
- B.Eng., Software Engineering. GPA: 3.6/4.0

Prizes and Awards

- **Excellent Degree Dissertation of South China University of Technology** 2023
- **Honorable Mention** in Mathematical Contest in Modeling 2023
- **National Scholarship** 2022
- ***Bronze Medal (46th)*** in ICPC Asia-East Continent Final(Xi An) 2022
- ***101/1608*** in CCF-DBCI Competition of "Small Sample Data Classification" 2022
- ***Silver Medal (46th)*** in ICPC Asia Regional Contest(Ji Nan) 2021
- ***44/3567*** in CCF-DBCI Competition of "Recognition of figure skaters' skeleton points based on Paddle" 2021
- ***First Prize*** in National Olympiad in Informatics in Province(NOIP) 2017

PrePrints and Publications

- *Wenqi Pei, Shizheng Hou, Boyan Li, **CHEN Han**, Zhichao Shi, Yuyu Luo.* **ROSE: An Intent-Centered Evaluation Metric for NL2SQL**. Accepted at ACL 2026 Main 2026
- ***CHEN Han**, Tao Huang, Phuong Pham, Shuang Wu.* **HiAE: A High-Throughput Authenticated Encryption Algorithm for Cross-Platform Efficiency** 2025
- ***CHEN Han**, Zicong Jiang, Zining Zhang, Bingsheng He.* **LogQuant: Log-Distributed 2-Bit Quantization of KV Cache with Superior Accuracy Preservation**. Accepted at ICLR 2025 Workshop on Sparsity in LLMs 2025
- *Wenqi Pei, Hailing Xu, Henry Hengyuan Zhao, **CHEN Han**, Zining Zhang, Shizheng Hou, Luo Pingyi, Bingsheng He.* **Optimizing Small Language Models for NL2SQL**. Accepted at ICLR 2025 Third Workshop on Deep Learning for Code 2025

Project Experience

- ***Cryptography Engineer: Efficient Cipher Implementation and Whitebox Design*** in **SG Digital Trust Lab, Singapore Research Center, 2012 Laboratory**
  Mentor: Dr. WU Shuang Jan.-June 2025
  - Design whitebox implementation of AES for different platforms, products and scenarios.
  - Implement HiAE cipher for kernel and user space, integrate it into several products, including mobile phone, server and auto-driving chips.
  - Optimize the performance of SM2, SM3 and SM4 ciphers on ARM platforms.

- ***Research Assistant: optimization for large language Model inference*** in **National University of Singapore**
  Mentor: Prof. HE Bingsheng May-Sept. 2024
  - Design a new 2-bit KV Cache quantization for LLMs base on attention patterns. achieve over 200% accuracy improvement at same compression rate.
  - Implement a adaptive API for popular inference frameworks like Python' s `transformers`, boosts batch size by 60% without increasing memory consumption.
  - Accepted at ICLR 2025 Workshop on Sparsity in LLMs. [Paper] [Code]
- ***Internship: Cryptography Engineer*** in **SG Digital Trust Lab, Singapore Research Center, 2012 Laboratory**
  Mentor: Dr. HUANG Tao Jan.-Dec. 2024
  - Design a *'XAXX'* structure to efficiently utilize the distinct pipelines of both ARM and x86 (with AES-NI) architectures, achieving high IPC.
  - Build a new AEAD (Authenticated Encryption with Associated Data) cipher named *'HiAE'* based on the *'XAXX'* structure, which is 5× faster than AES-256-GCM across different platforms and outperforms all existing AEAD ciphers on latest ARM and x86 processors.
  - Create a new record of 340Gbps throughput for single-threaded and single-stream AEAD encryption. Applied to the various products.
  - Open-source on Cryptology ePrint Archive [Paper] [Code]
- ***Research Assistant: Parallel Graph Partitioning*** in **Hong Kong University of Science and Technology, Guangzhou**
  Mentor: Prof. Zeyi WEN Apr-Sept. 2023
  - Implement a library of parallel hyper graph partitioning with openMP task.
  - Realize a multiple node contraction algorithm for graph coursenning.
  - Research parallel graph partitioning algorithm and adapt it for fill-in reduction of sparse matrix.
- ***Symmetric Matrix Solving Algorithm Parallel Optimization for ARM Architecture***
  Mentor: Prof. TANG Deyou May-Dec. 2022
  - Optimize and parallel Bounded Bunch-Kaufman Algorithm(*sysv_rk subroutine of LAPACK) for solving symmetric matrix on ARM server processor with NEON instruction set and openMP.
  - Implement a parallel column reordering method in row swap of solving symmetric matrix to enhance memory access locality for column major matrix for better cache hit rate and parallelism, achieving a performance improvement from 320Gflops to 580Gflops.
  - Implement the same optimization on Skylake Intel processor and achieve 2-5x multi-core speedup than MKL library for *sytrs_3 subroutine of LAPACK.
  - Awarded as the Excellent Degree Dissertation of South China University of Technology.

Technical Skills

- *English*: IELTS(6.5), CET-4, CET-6.
- *Programming Languages*: C/C++, Fortran, p4-16, Python, SQL, LaTeX.
- *Technical Skills*: openMP, SIMDs(NEON, AVX512), MPI, PyTorch, CUDA.
- *TestDemo Certificate*: C++, TOP 10%, LINUX, TOP 10%, PYTHON, TOP 10%.
- *Kaggle Certificate*: Data Visualization, Intro to Machine Learning, Intro to Deep Learning, Intro to Game AI and Reinforcement Learning.

- **Online Academic Program on Machine Learning, McGill University** Jan.-Feb. 2022

Exchange Experience

陈涵

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联系方式

Tel: +86-13305014345
Tel: +65-89420214
微信: concyclics
✉ E-mail:chenhan@u.nus.edu
Github:www.github.com/Concyclics
Address: NUS School of Computing, 13 Computing Drive, Singapore 117417

教育经历

新加坡国立大学, 新加坡 — 2025–现在
- 计算机科学博士. — 导师: 何丙胜教授

新加坡国立大学, 新加坡 — 2023–2025
- 计算机科学硕士. — GPA: 4.4/5.0

华南理工大学, 广东省广州市 — 2019–2023
- 工学学士, 软件工程专业. — GPA: 3.6/4.0

获奖荣誉

- **华南理工大学本科优秀毕业设计** *(论文)* — 2023
- **二等奖**美国大学生数学建模竞赛 (MCM/ICM) — 2023
- **铜牌**第 46 届 ICPC 国际大学生程序设计竞赛亚洲区决赛 — 2022
- ***101/1608*** CCF-DBCI ” 小样本数据分类算法” 竞赛 — 2022
- **国家奖学金** — 2022
- **银牌**第 46 届 ICPC 国际大学生程序设计竞赛 (济南站) — 2021
- ***44/3567*** CCF-DBCI ” 基于飞桨实现花样滑冰选手骨骼点识别” 竞赛 — 2021
- **一等奖**全国青少年信息学奥林匹克联赛 (NOIP) — 2017

预印本和发表论文

- *Wenqi Pei, Shizheng Hou, Boyan Li, **CHEN Han**, Zhichao Shi, Yuyu Luo.* **ROSE: An Intent-Centered Evaluation Metric for NL2SQL**. 录用于 ACL 2026 Main — 2026
- ***CHEN Han**, Tao Huang, Phuong Pham, Shuang Wu.* **HiAE: A High-Throughput Authenticated Encryption Algorithm for Cross-Platform Efficiency** — 2025
- ***CHEN Han**, Zicong Jiang, Zining Zhang, Bingsheng He.* **LogQuant: Log-Distributed 2-Bit Quantization of KV Cache with Superior Accuracy Preservation**. Accepted at ICLR 2025 Workshop on Sparsity in LLMs — 2025
- *Wenqi Pei, Hailing Xu, Henry Hengyuan Zhao, **CHEN Han**, Zining Zhang, Shizheng Hou, Luo Pingyi, Bingsheng He.* **Optimizing Small Language Models for NL2SQL**. Accepted at ICLR 2025 Third Workshop on Deep Learning for Code — 2025

项目经历

- **密码算法工程师*:* 高效密码算法实现和白盒设计**: 新加坡数字信任实验室, 华为 2012 实验室新加坡研究所 — 2025/01–2025/06
  导师: 吴双博士
  - 设计了不同场景和平台的 AES 白盒实现。
  - 在用户态和内核态实现了 HiAE 密码算法, 并集成到多种产品中, 包括手机、服务器和自动驾驶芯片。
  - 优化了 SM2、SM3 和 SM4 密码算法在 ARM 平台上的性能。
- **科研助理*:* 大语言模型推理优化**: 新加坡国立大学 — 2024/05–2024/09
  导师: 何丙胜教授

- 设计了一种基于注意力模式的 2 位 KV Cache 量化方法，在相同压缩率下，提高了 200% 的准确率。
- 实现了一个适应性 API，用于流行的推理框架，如 Python 的 transformers，在不增加内存消耗的情况下，将批处理大小提高了 60%。
- 该项目已被 ICLR 2025 Sparsity in LLMs Workshop 接受。[Paper] [Code]

- **实习生：密码算法工程师**: 华为 2012 实验室新加坡研究所谢尔德实验室 2024/01–2024/12
  导师：黄涛博士
  - 设计了一种新的 *'XAXX'* 算法结构，可以高效利用 ARM 和 x86(带 AES-NI) 架构的流水线，实现了高 IPC。
  - 基于 *'XAXX'* 结构构建了一种新的 AEAD(带关联数据的认证加密) 密码算法，*'HiAE'*，在多种平台上相较 AES-256-GCM 提升 5 倍以上性能，在最新的 ARM 和 x86 处理器上优于所有现有的 AEAD 密码算法。
  - 创造了单线程单流 AEAD 加密的 340Gbps 新纪录，并应用于多种华为产品。
  - 该项目已在 Cryptology ePrint Archive 上开源。[Paper] [Code]

- **科研助理**: 香港科技大学广州校区 2023/04–2023/09
  导师：文泽忆教授
  - 实现了一个基于 openMP Task 的并行多层拓扑图分割库。
  - 在图压缩中实现了一种多节点收缩算法。
  - 研究将并行图分割算法，应用于稀疏矩阵的填充减少。

- **对称矩阵函数求解 *BBK* 算法的并行优化** 2022/04-2022/12
  导师：汤德佑教授
  - 在 ARM 处理器上利用 NEON 指令集和 openMP 对 Bounded Bunch-Kaufman 算法 (LAPACK 库 *sysv_rk 函数) 进行并行优化。
  - 实现了一种并行列重排方法，在列优先矩阵的行交换中改进访存局部性，使得缓存命中率和并行性能得到提高，在鲲鹏 920-6426 处理器上的单精度性能从 320Gflops 提升到 580Gflops。
  - 将该方法移植到 Intel Skylake 处理器上，对比 MKL 库的 *sytrs_3 函数，实现了 2-5 倍的并行性能提升。
  - 该项目获评华南理工大学本科优秀毕业设计。

专业技能

- 英语认证水平：CET-4, CET-6, IELTS(6.5).
- 编程语言：C/C++, Fortran, p4-16, Python, SQL, LaTeX.
- 编程技能：openMP, SIMDs(NEON, AVX512), MPI, PyTorch, CUDA.
- *TestDemo* 编程技能认证：C++, TOP 10%, LINUX, TOP 10%, PYTHON, TOP 10%
- *Kaggle* 课程认证：数据可视化，机器学习，深度学习，强化学习

交换经历

- **机器学习线上访学项目，麦吉尔大学** 2022/01–2022/02